NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Batch Number	BG-10
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Guide	M.Sireesha
Title	WasteNet: A Structured-Pruned and EMA-Augmented RepVGG Framework for Real-Time Trash Classification
Domain/Technology	DEEP LEARNING
Base Paper Link	https://ieeexplore.ieee.org/document/10900382
Dataset Link	https://www.kaggle.com/asdasdasasdas/garbage-classification https://www.kaggle.com/garythung/trashnet https://www.kaggle.com/techsash/waste-classification-data
Software Requirements	Browser: Any latest browser like Chrome Operating System: Windows 7 Server or later Python (COLAB)
Hardware Requirements	SystemType: Intel Core i5 or above RAM: 8 GB Number of cores:5 Number of Threads: 4
Abstract	Achieving efficient classification of garbage boosts recycling and helps with pollution control as well as sustain able waste management. To address this challenge,we developed RepVGG-MEM, which is a lightweight and highly accurate Deep Learning model for real-time waste classification. Built on RepVGG, the model applies EMA,Mixup data augmented training, and structured pruning to optimization to maintain a strong speed performance ratio. The model and its described metrics, performance of 95.06% accuracy and a 94.5% F1 score, even with closely resembling waste categories surpass several previous versions. Moreover, its accuracy makes it ideal for edge devices and smart mobile waste management systems.RepVGG MEM shows better performance and generalization than earlier versions. Its lightweight design makes it easy to deploy on edge devices such as smart bins and mobile platforms. This capability positions it as a promising solution for automated waste management in real-world situations.

Signature of the student(s) **Signature of the Guide** **Signature of the project coordinator**